

GOPICHAND HINGE

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Senior Generative AI Engineer

PROFESSIONAL SUMMARY

- Senior **Generative AI** and **ML Engineer** with 10+ years of experience designing and deploying enterprise AI solutions across financial, healthcare, retail, and public sectors, specializing in **GenAI systems** including **RAG pipelines** and **agentic workflows**.
- Built production-grade **LLM applications** using **AWS Bedrock (Claude)**, while leveraging OpenAI GPT-4 and Azure OpenAI for prototyping and evaluation, integrating hybrid retrieval with vector search (Pinecone, FAISS) to enable low-latency semantic search.
- Designed and implemented **agentic AI systems** using **LangGraph** and **tool-calling frameworks**, enabling autonomous multi-step reasoning, dynamic decision-making, and orchestration of external tools within production-grade GenAI workflows.
- Developed advanced **RAG and agentic pipelines** using **LangChain**, **LangGraph**, and **Hugging Face Transformers**, enabling multi-step reasoning, tool orchestration, and dynamic workflow execution for complex enterprise use cases.
- Optimized LLM performance** through **prompt engineering (CoT, ReAct)**, retrieval tuning, and **parameter-efficient fine-tuning (LoRA/QLoRA)**, improving response accuracy, reducing latency, and optimizing token-level inference costs in production systems.
- Deployed cloud-native **AI systems on AWS** and **Azure** using **Bedrock** and **S3**, integrating robust **MLOps** and **DevOps pipelines** for scalable model deployment, real-time monitoring, and end-to-end lifecycle management across production environments.
- Engineered scalable **data pipelines** using **Python, SQL, Spark, Databricks, Airflow, and Delta Lake**, integrating structured, semi-structured, and streaming data sources to support large-scale ML and GenAI workloads.
- Built and validated **machine learning models** using regression, classification, clustering, and time-series techniques, applying feature engineering and statistical modeling to solve domain-specific problems across large-scale datasets.
- Applied **data science methodologies** including hypothesis testing, statistical analysis, and **A/B experimentation** to evaluate model performance, validate assumptions, and drive data-driven decision-making in production environments.
- Developed scalable **backend systems and APIs** using **Python (FastAPI, Flask)**, implementing modular architectures and testing frameworks to ensure reliability, maintainability, and production readiness of AI services.

Technical Skills

Programming Languages	Python (Pandas, NumPy, SciPy, Scikit-learn, PyTorch), SQL, R, Java, JavaScript, TypeScript, Bash
Generative AI, LLMs & Retrieval	AWS Bedrock, Anthropic Claude, OpenAI (GPT-4, GPT-4o), RAG, Hybrid Retrieval, Prompt Engineering (CoT, ReAct, Tool Calling), Agentic Workflows, LoRA/QLoRA, LangChain, LangGraph, Hugging Face Transformers, Pinecone, FAISS, Embeddings, Vector Databases, Vector Search, Guardrails (Hallucination Control, Safety Filtering, Output Validation)
Machine Learning, NLP & Evaluation	Regression, Classification, Clustering, Random Forest, Gradient Boosting, XGBoost, LightGBM, K-Means, ARIMA, Natural Language Processing, Feature Engineering, AUC, KS, Gini, Precision/Recall, F1-Score, NDCG, MAP, MRR, Cross-Validation, PSI, A/B Testing, SHAP, LIME
Data Engineering, Big Data & Storage	Apache Spark (PySpark), Databricks, Delta Lake, Apache Airflow, Apache Kafka, ETL/ELT Pipelines, Batch & Streaming Processing, Lakehouse Architecture, Distributed Systems, PostgreSQL, SQL Server, Oracle, MongoDB, DynamoDB, Amazon Redshift
Backend, APIs & Microservices	FastAPI, Flask, REST APIs, Microservices Architecture, API Performance Optimization
Cloud Platforms	AWS (S3, EC2, EKS, Lambda, Redshift, CloudWatch, Bedrock), Azure (Azure OpenAI – prototyping/evaluation, Azure ML – exposure)
DevOps, MLOps, LLMOps & Observability	Docker, Kubernetes, Terraform, MLflow, CI/CD (Jenkins, GitHub Actions), Prometheus, CloudWatch, Drift Detection, Latency Optimization, Experiment Tracking, Model Registry, Kubeflow
Data Visualization & BI	Streamlit, Tableau, Power BI, Matplotlib, Plotly
Tools, Practices & Governance	Git, Linux, Jupyter Notebook, Agile/Scrum, Model Governance, HIPAA Compliance

PROFESSIONAL EXPERIENCE

Client: Starbucks, Seattle, WA

MAY 2023 - CURRENT

Role: Senior Generative AI Engineer

Responsibilities:

- Designed and deployed an end-to-end **Generative AI personalization platform** using **AWS Bedrock (Claude), LangChain**, and **Pinecone** on **AWS EKS** and **Databricks Lakehouse**, improving retrieval accuracy by 35% and reducing response latency by 30%.
- Designed a **cloud-native microservices architecture** using **AWS EKS, Apache Kafka, and Delta Lake**, enabling real-time and batch data processing with scalable, fault-tolerant, and low-latency performance for AI applications.
- Worked in iterative, **sprint-based delivery cycles**, partnering with product owners, UX teams, and data engineers to translate business requirements into scalable **RAG pipelines, personalization APIs, and deployable AI services**.
- Built real-time and batch ingestion pipelines using **Kafka, REST APIs, and Databricks Autoloader**, integrating POS transactions, loyalty data, mobile activity, and enterprise documents into **Delta Lake** for downstream ML and GenAI processing.
- Performed large-scale data transformation and feature engineering using **PySpark and Pandas**, enabling session aggregation, behavioral scoring, and data enrichment to support recommendation models and **RAG-based retrieval systems**.
- Architected optimized **Delta Lake storage** with partitioning and indexing strategies, enabling efficient feature engineering, model training, and scalable data access for AI workloads while maintaining ACID compliance and schema evolution.
- Built scalable **vector search systems** using **Pinecone for low-latency production workloads** and **FAISS for offline evaluation**, integrating embeddings into semantic retrieval pipelines for **RAG-based systems**.
- Selected and benchmarked **AWS Bedrock (Claude)** models based on reasoning capability, latency, and cost, while evaluating **OpenAI GPT-4 and Azure OpenAI** for prototyping and benchmarking, standardizing production workloads on **AWS Bedrock**.
- Designed and implemented advanced **RAG pipelines** using **Top-K retrieval, cross-encoder re-ranking, and query understanding techniques**, improving retrieval relevance and robustness across diverse production-scale data environments.
- Optimized LLM performance using **prompt engineering techniques (CoT, ReAct)**, retrieval tuning, token management, and response caching strategies, improving response accuracy, latency, and efficiency across high-throughput production systems.
- Developed **agentic orchestration pipelines** using **LangChain and LangGraph**, enabling multi-step reasoning, tool calling, state management, and dynamic execution of complex workflows across distributed LLM-driven applications.
- Built **agentic tool integration frameworks** enabling **LLMs** to interact with **external APIs**, knowledge sources, and business services for dynamic task execution and context-aware decision-making in production systems.
- Applied modular **object-oriented Python design** and clean architecture principles to develop reusable service layers, improving maintainability, scalability, and separation of concerns across distributed AI systems and microservice architecture.
- Conducted **A/B testing and offline evaluation** using retrieval precision, response relevance, and faithfulness metrics, **validating LLM performance** and ensuring production readiness of **Bedrock-based systems**.
- Containerized inference services using **Docker** and deployed on **Kubernetes (AWS EKS)** with autoscaling, supporting high-throughput, **low-latency AI workloads** in production environments.
- Built and maintained **CI/CD pipelines** using **Jenkins and GitHub Actions**, automating model deployment, API releases, and environment updates across development, staging, and deployment environments.
- Provisioned **infrastructure** using Terraform and **tracked experiments** with MLflow, enabling reproducible environments, version control, and efficient lifecycle management of AI and machine learning workloads with automated workflows.
- Implemented monitoring and observability using **CloudWatch** and **Prometheus**, tracking API latency, model behavior, response relevance, and **hallucination metrics** to ensure consistent **LLM performance** in production.
- Developed **unit and integration tests using PyTest**, including API validation, prompt testing, and pipeline checks, ensuring data flow accuracy, service reliability, and inference correctness to support UAT and production readiness in live environments.
- Created **API documentation** using **Swagger/OpenAPI** and maintained **Confluence knowledge bases**, enabling knowledge sharing, onboarding, and long-term maintainability of systems and services across teams and stakeholders.

Environment: Python, SQL, PySpark, Databricks, Delta Lake, Apache Kafka, Apache Airflow, AWS Bedrock (Claude), Hugging Face Transformers, LangChain, LangGraph, Redis, Prometheus, CloudWatch, Pinecone, FAISS, AWS (S3, EKS, Lambda, API Gateway, IAM), Docker, Kubernetes, Terraform, MLflow, Jenkins, GitHub Actions, FastAPI, Flask, PostgreSQL, MongoDB, Tableau, Power BI, Git, OpenAI (GPT-4 - POC/Evaluation)

Client: Citigroup - New York, NY

Mar 2020 – Apr 2023

Role: Senior Machine Learning Engineer (Risk Analytics & Data Engineering)

Responsibilities:

- Developed and deployed **credit risk and delinquency prediction models** using **Python, PySpark, and Scikit-learn** on AWS infrastructure, improving early delinquency detection by **18%** and strengthening portfolio risk decisioning across lending systems
- Designed **scalable data lakehouse architecture** integrating data lake and warehouse environments using **Spark, SQL, and AWS (S3, EC2)** for distributed data processing and model training and **high-volume model training pipelines**.
- Worked in **Agile/Scrum (2-week sprints)**, collaborating with risk, finance, compliance, and product teams to translate regulatory requirements into measurable KPIs, validation frameworks, and production-ready risk models and solutions.
- Built **data ingestion pipelines** integrating structured and semi-structured data from core banking systems using **Python, SQL, and Apache Spark**, ensuring reliable data availability for downstream modeling and analytics workflows across distributed systems.
- Performed large-scale data cleaning, transformation, and validation using **PySpark and Pandas** in live environments, implementing data quality checks to ensure accurate model training and regulatory compliance across diverse enterprise datasets.
- Optimized SQL queries and designed scalable storage and feature datasets across **Oracle, SQL Server, and S3**, improving processing efficiency and enabling large-scale analytics and model training workflows for **high-volume data processing**.
- Engineered scalable **feature pipelines and feature datasets** (utilization ratios, delinquency buckets, EAD), ensuring consistency, reusability, and data integrity across **production-grade risk modeling workflows** in distributed environments.
- Built and validated **probability of default (PD) and early delinquency models** using **Logistic Regression, Decision Trees, and XGBoost**, applying calibration and risk segmentation techniques to improve interpretability and regulatory alignment
- Optimized **model performance** using hyperparameter tuning, feature selection, and cross-validation, improving model stability, generalization, and robustness across diverse customer segments and evolving **data distributions** in large-scale systems
- Executed model validation using **AUC, KS, Gini, and PSI metrics**, ensuring compliance with model risk management (**MRM**) frameworks and supporting regulatory standards for credit risk modeling and governance across production risk environments.
- Managed **model lifecycle using MLflow**, including experiment tracking, versioning, and controlled deployment of machine learning models across development and production environments with **governed model lifecycle management**.
- Built and optimized batch scoring pipelines using **Apache Spark** on AWS (**EC2, S3**), improving inference latency and throughput for large-scale portfolio scoring and enterprise analytics integration with **efficient distributed compute execution**.
- Built distributed **ML pipelines** on AWS-integrated infrastructure, enabling scalable model training and batch scoring across high-volume datasets, and enterprise risk analytics systems with **parallelized data processing frameworks**.
- Packaged and containerized **machine learning models using Docker**, enabling consistent deployment and scalable batch inference across distributed systems with portable execution layers supporting **efficient multi-environment deployment workflows**.
- Implemented CI/CD pipelines using Jenkins, automating model packaging, versioning, validation, and deployment workflows for reliable and repeatable ML releases with **end-to-end pipeline automation workflows**.
- Established **monitoring frameworks** to track **feature drift, prediction stability, and population shifts** in production pipelines. Enabled continuous evaluation of model performance and supported proactive model recalibration when needed.
- Developed **unit and integration testing frameworks using PyTest** and supported **UAT validation cycles**, ensuring correctness and reliability of model outputs and scoring pipelines with **comprehensive testing and validation coverage**.
- Documented model logic, validation methodologies, and pipeline workflows while managing **version-controlled ML lifecycle using Git and MLflow**, ensuring audit readiness, reproducibility, and regulatory compliance.

Environment: Python (Pandas, NumPy, SciPy, Statsmodels, Scikit-learn), R, SQL, Apache Spark (PySpark), Oracle, SQL Server, PostgreSQL, AWS (S3, EC2), Tableau, Power BI, Jupyter Notebook, Git, Linux

Client: L.A. Care Health Plan, Los Angeles, CA

Dec 2017 – Feb 2020

Role: Machine Learning Engineer

Responsibilities:

- Developed end-to-end machine learning solutions for **patient risk stratification** and **population health analytics** using Python and Scikit-learn, enabling data-driven prioritization and decision-making across large-scale healthcare platforms.
- Designed scalable data architecture integrating clinical and case management **healthcare datasets** using **SQL-based systems**, enabling efficient data processing, risk modeling, and resource allocation across distributed analytics workflows.

- Worked in **Agile/Scrum (2-week sprints)**, collaborating with healthcare stakeholders, policy teams, and compliance analysts to translate business and regulatory requirements into scalable, production-ready predictive models and analytics solutions.
- Built batch data ingestion pipelines integrating **claims, case management, and healthcare datasets** using Python, SQL, and Apache Airflow, ensuring reliable, consistent, and timely data availability for downstream analytics and machine learning processes.
- Performed large-scale data preprocessing, cleansing, validation, and **feature engineering** using Pandas and NumPy, improving data quality and preparing structured, high-quality datasets for predictive modeling and advanced analytics workflows.
- Designed scalable data storage solutions using **PostgreSQL and Amazon Redshift**, enabling high-performance querying, historical data analysis, and efficient execution of large-scale machine learning and reporting pipelines.
- Engineered feature pipelines capturing **patient behavior, healthcare utilization patterns, and socioeconomic indicators**, enabling robust feature sets for risk modeling, segmentation, and population-level healthcare analytics.
- Built predictive models for **patient risk scoring, readmission likelihood, and healthcare demand forecasting** using Logistic Regression, Random Forest, XGBoost, and LightGBM, supporting proactive decision-making and resource planning.
- Conducted **exploratory clinical data analysis (EDA)** and statistical analysis to identify key drivers influencing patient outcomes, utilization trends, and healthcare service demand patterns across diverse datasets.
- Applied **hypothesis testing and A/B experimentation** for care interventions and program effectiveness, enabling data-driven optimization of healthcare strategies and improving operational efficiency and patient outcomes.
- Optimized model performance using **hyperparameter tuning, feature selection, and cross-validation**, improving model accuracy, generalization, and robustness across evolving healthcare datasets and real-world scenarios.
- Implemented NLP pipelines using **spaCy and NLTK** to process unstructured clinical notes and case management data, extracting meaningful insights for prioritization, classification, and decision support applications.
- Evaluated models using **ROC-AUC, precision, recall, and F1-score**, ensuring model reliability, interpretability, and alignment with business objectives and healthcare analytics requirements.
- Deployed machine learning models as **REST APIs** using Flask and Docker on AWS (EC2, S3), enabling scalable, low-latency inference and seamless integration with downstream healthcare analytics systems.
- Optimized inference performance by tuning **API response times and batch processing workflows**, improving system throughput, latency, and scalability across large-scale machine learning pipelines.
- Implemented **CI/CD pipelines using Jenkins**, automating model deployment, versioning, testing, and reproducibility across development and production environments for consistent delivery.
- Managed **cloud-based workflows and infrastructure on AWS**, supporting scalable machine learning pipelines, distributed data processing, and efficient resource utilization across environments.
- Established monitoring frameworks to track **model performance, data drift, and prediction stability**, enabling continuous evaluation, alerting, and proactive model maintenance in production systems.
- Developed validation checks, supported UAT cycles, and maintained documentation ensuring **correctness, audit readiness, healthcare data governance (HIPAA-aligned), and long-term system maintainability**.

Environment: Python (Pandas, NumPy, Scikit-learn), SQL, Apache Airflow, PostgreSQL, Amazon Redshift, AWS (EC2, S3), Flask, Docker, Jenkins, spaCy, NLTK, Tableau, Power BI, Git, Linux, XGBoost, LightGBM

Client: Apple, Sunnyvale, CA

May 2016 - Nov 2017

Role: Data Scientist

Responsibilities:

- Built scalable **data science solutions** to analyze large-scale product, operational, and documentation datasets using Python and SQL, enabling improved analytics, search, and classification across engineering and business teams.
- Designed end-to-end **analytical pipelines** processing structured and unstructured data, supporting downstream machine learning, reporting, and decision-support systems across product and operations domains.
- Worked in **Agile/Scrum environments**, collaborating with engineering, product, and analytics teams to translate functional requirements into scalable, data-driven analytical solutions across large-scale data platforms.
- Integrated and processed data from multiple internal systems using **Python and SQL**, performing data cleaning, validation, and preprocessing with Pandas and NumPy to create high-quality, reusable datasets for analytics and ML workflows.
- Designed and maintained **data warehouse schemas (Star and Snowflake models)**, supporting historical analysis, reporting, and scalable analytical workloads across distributed data environments.
- **Engineered feature pipelines** and applied dimensionality reduction (**PCA**) to improve data quality and enable efficient downstream model training and analytics workflows across large-scale machine learning systems.

- Developed supervised and unsupervised machine learning models using **Logistic Regression, Decision Trees, Random Forest, K-Means, and XGBoost** for classification, clustering, and anomaly detection use cases.
- Built NLP pipelines using **NLTK with TF-IDF, tokenization, and similarity scoring**, enabling document classification, semantic search, and improved content organization across large-scale document processing systems.
- Developed deep learning models using **TensorFlow and Keras**, optimizing performance through feature engineering, hyperparameter tuning, and validation techniques to improve model accuracy and generalization.
- Deployed machine learning models as **Flask-based REST APIs**, containerized with Docker and hosted on **AWS (EC2, S3, Redshift)** to enable scalable analytics and inference.
- Supported model validation and testing workflows to ensure accuracy, consistency, and reliability of machine learning outputs across analytical systems.
- Maintained version-controlled workflows using **Git** and documented analytical processes to ensure reproducibility and long-term maintainability.

Environment: Python, SQL, Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Keras, NLTK, Matplotlib, AWS (EC2, S3, Redshift), Flask, Power BI, Data Warehousing (Star & Snowflake Schema), PCA, Classification, Clustering, Regression, Git

Client: Apollo Hospitals – Chennai, India

July 2013 to Feb 2016

Role: Data Scientist / Data Analyst

Responsibilities:

- Built **hospital operations and revenue-cycle analytics solutions** using Python and SQL, analyzing OPD/IPD volumes, bed occupancy, and billing trends to enable data-driven decision-making across clinical and administrative teams.
- Designed scalable **analytical workflows** for operational efficiency, capacity planning, and financial performance monitoring, supporting decision-making across multiple hospital departments and high-volume healthcare environments.
- Collaborated with **clinicians, finance teams, and operations stakeholders** to translate healthcare requirements into actionable analytics, KPIs, and reporting solutions aligned with hospital performance goals across enterprise healthcare delivery systems.
- Developed end-to-end **ETL pipelines** using Python and SQL to integrate data from EMR, HIS, billing, laboratory, and pharmacy systems into centralized analytics repositories for downstream processing across integrated hospital data platforms.
- Performed data cleansing, normalization, validation, and **feature engineering** on clinical and financial datasets, ensuring accuracy, consistency, and reliability for analytics and machine learning workflows in regulated healthcare data environments.
- Designed and maintained **relational data models** in SQL Server and PostgreSQL, enabling scalable reporting, historical analysis, and efficient data retrieval across healthcare systems and data platforms.
- Built predictive models using **ARIMA, Regression, Random Forest, Gradient Boosting, and XGBoost**, applying statistical modeling and segmentation techniques to forecast patient inflow and resource utilization.
- Optimized model performance using **feature engineering, hyperparameter tuning, and validation techniques**, improving accuracy, stability, and generalization across diverse clinical and financial datasets.
- Developed **Tableau and Power BI dashboards** and performed cost-of-care and revenue analysis, providing visibility into operational metrics such as LOS, bed turnover, and revenue trends.
- Implemented data validation and quality checks to ensure accuracy and consistency of analytics outputs, supporting reliable decision-making across hospital operations and financial reporting systems.
- Supported validation workflows to ensure correctness and reliability of predictive models and analytics outputs.
- Maintained documentation and version-controlled workflows using Git and Jupyter, ensuring reproducibility, audit readiness, and long-term maintainability of healthcare analytics solutions.

Environment: Python (Pandas, NumPy, SciPy, Scikit-learn), R, SQL, XGBoost, SQL Server, PostgreSQL, Tableau, Power BI, Jupyter Notebook, Time-Series (ARIMA), Regression, Classification, Clustering, Data Warehousing, Git, Linux

EDUCATION: PARUL UNIVERSITY - Bachelor of Technology (Computer Science)